COMP0087

Statistical Natural Language Processing

Lecture 13 – Evaluation

Slides present original content as well as content based on what was previously developed by:

- Stanford CS224N

Two questions:

- What to evaluate on?
  - Benchmarks (not our focus)
- How to evaluate?
  - Today's lecture

Model-free or model-based metrics

Reference-based or reference-free metrics

LLM as a judge

Data Contamination

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	
QA with a sentence answer	
QA with long-form answers	

Examples

Question	Correct Answer
What is the capital of Iran? A) Berlin B) Madrid C) Tehran D) Rome	C
Which planet is closest to the Sun? A) Earth B) Mercury C) Venus D) Mars	B
What is 12 × 8? A) 96 B) 84 C) 108 D) 72	A

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	Text span matching or exact numeric/expression match
QA with a sentence answer	
QA with long-form answers	

Examples

Question	**Correct Answer**
Who invented the telephone?	Alexander Graham Bell
What is the square root of 144?	12

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	Text span matching or exact numeric/expression match
QA with a sentence answer	➢ ???
QA with long-form answers	

Examples

Question	Why do rabbits have long ears?
Correct Answers	**Answer A**: "Rabbits have long ears to maximize sound collection and detect predators from great distances, and also to regulate body temperature by dissipating heat." **Answer B:** "Long ears help rabbits hear really well and keep cool in hot weather."

Both are arguably correct.
Exact match is useless here.

How to evaluate

Answer type	Grading complexity
Multiple-choice QA	straightforward accuracy (easiest!)
QA with a short answer	Text span matching or exact numeric/expression match
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Model-free or model-based metrics

Reference-based or reference-free metrics

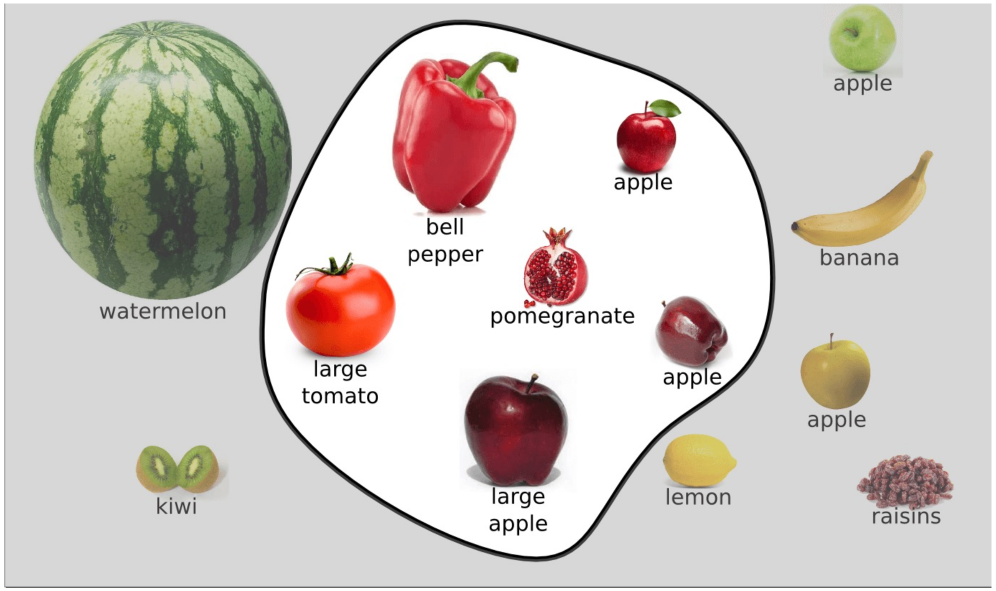
LLM as a judge / jury

Classical metrics are model-free metrics

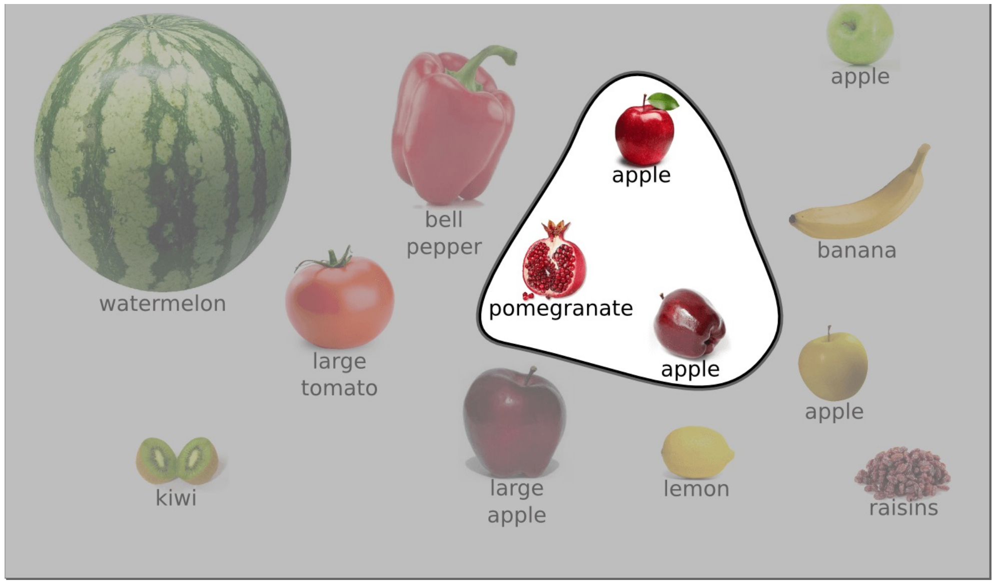
BLEU	**Machine Translation**	Precision-focused; measures n-gram overlap.	**Higher is better (1 or 100)**
ROUGE	**Summarization**	Recall-focused; measures n-gram overlap.	**Higher is better (1)**
METEOR	**MT & Dialogue**	Includes stemming and synonymy (more semantically aware); unigram-based.	**Higher is better (1)**
TER	**Translation Quality**	Measures the Edit Distance	**Lower is better (0)**
WER	**Speech Recognition**	Standard Word Error Rate	**Lower is better (0)**

Side note – in case you don't know Precision, Recall

Reference: You were looking for these



Candidate: You retrieved these



Precision = 3/3 = 1

Recall = 3/6 = 0.5

F-measure = $2\times\frac{1\times0.5}{1+0.5} = 0.66$

Question?

Why precision = 1 or Recall = 1

Could be problematic?

Classical metrics are model-free metrics

BLEU (Papineni et al., 2002)

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\Big(\sum_{n=1}^{N} w_n \log p_n\Big)$$

$$p_n = \frac{\sum_{\text{n-gram} \in c} \min\big(\mathrm{Count}(\text{n-gram}, c),\ \mathrm{Count}(\text{n-gram}, \mathrm{ref})\big)}{\sum_{\text{n-gram} \in c} \mathrm{Count}(\text{n-gram}, c)}$$

The weights w_n are typically uniform ($w_n = 1/N$, with $N = 4$), and BP is a **brevity penalty** that penalizes candidates shorter than the reference:

$$\mathrm{BP} = \begin{cases} 1 & \text{if } |c| > |r| \\ \exp(1 - |r|/|c|) & \text{if } |c| \le |r| \end{cases}$$

where $|c|$ and $|r|$ are the lengths of the candidate and reference.

The cat is sitting on the mat

Unigrams (n = 1)

The, cat, is, sitting, on, the, mat

Bigrams (n = 2)

The cat
cat is
is sitting
sitting on
on the
the mat

Trigrams (n = 3)

The cat is
cat is sitting
is sitting on
sitting on the
on the mat

4-grams (n = 4)

The cat is sitting
cat is sitting on
is sitting on the
sitting on the mat

Classical metrics are model-free metrics

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where $|c|$ and $|r|$ are the lengths of the candidate and reference.

Ref: The cat is sitting on the mat
Candidate: The cat

Length |ref| = 7 Length |candidate| = 2

Precision for **n=1** (unigram) is

RefU=[the, cat, is, sitting, on, the, mat]
CanU=[the, cat]

Count ("the", CanU)=1
Count ("cat", CanU)=1
Count ("the", RefU)=2
Count ("cat", RefU)=1

Precision = $\frac{1}{1+1}$(min(1,1)+min(1,2)) = $\frac{1}{1+1}(1+1) = 1$

Classical metrics are model-free metrics

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where $|c|$ and $|r|$ are the lengths of the candidate and reference.

Ref: The cat is sitting on the mat
Candidate: The cat

Length |ref| = 7 Length |candidate| = 2

Without BP, the BLEU1=1.

Because |c| < |r|, we apply:

BP=exp(1−|r|/|c|)
BP≈0.0821

BLEU1≈0.082

Classical metrics are model-free metrics

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where $|c|$ and $|r|$ are the lengths of the candidate and reference.

BLEU: n-gram count-based metric for machine translation to capture the closeness between the reference (gold-standard, usually human-written) text and the model generated output

Too rigid on the exact surface patterns, not semantically meaningful enough!

Classical metrics are model-free metrics

BLEU (Papineni et al., 2002)

$$\text{BLEU} = \text{BP} \cdot \exp\Big(\sum_{n=1}^{N} w_n \log p_n\Big)$$

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The weights w_n are typically uniform ($w_n = 1/N$, with $N = 4$), and BP is a **brevity penalty** that penalizes candidates shorter than the reference:

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where $|c|$ and $|r|$ are the lengths of the candidate and reference.

BLEU: 0.131
METEOR: 0.410

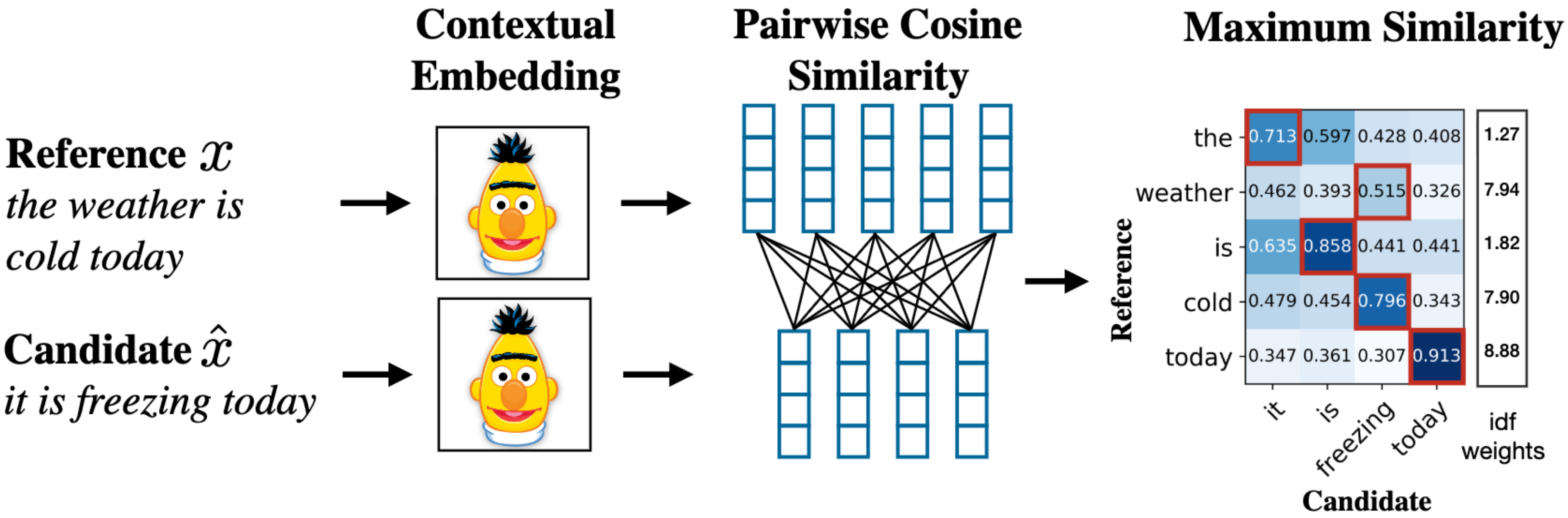
Ref: The dog sat on the mat.
Candidate: The puppy did sit on the rug.

BLEU: n-gram count-based metric for machine translation to capture the closeness between the reference (gold-standard, usually human-written) text and the model generated output

Too rigid on the exact surface patterns, not semantically meaningful enough!

Model-based metrics!

- BertScore (Zhang et al., 2020)
  - A soft, embedding version of BLEU / ROUGH



Model-based metrics!

- Vector similarity

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \; \|\mathbf{v}\|}$$

- BertScore (Zhang et al., 2020)
  - A soft, embedding version of BLEU / ROUGH

Ref: The dog sat on the mat.
Candidate: The puppy did sit on the rug.

BLEU: 0.13
METEOR: 0.41
BERTSCore-F: 0.96

BertScore

Given a reference $R = (r_1, \ldots, r_m)$ and candidate $C = (c_1, \ldots, c_n)$, first encode all tokens through a pretrained BERT model to get contextual embeddings. Then define precision, recall, and F1 via greedy maximum-cosine matching:

$$P_{\text{BERT}} = \frac{1}{|C|} \sum_{c_j \in C} \max_{r_i \in R} \cos(c_j,\; r_i)$$

$$R_{\text{BERT}} = \frac{1}{|R|} \sum_{r_i \in R} \max_{c_j \in C} \cos(r_i,\; c_j)$$

$$F_{\text{BERT}} = 2 \cdot \frac{P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

Model-based metrics!

- Vector similarity

$$\text{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}$$

- BertScore (Zhang et al., 2020)
  - A soft, embedding version of BLEU / ROUGH
- BLUERT (Pu et al., 2021)
  - A model trained to mimic human evals

BLUERT

BLEURT TRAINING PIPELINE

Phase 1 — Synthetic pretraining. Generate millions of (reference, perturbed) pairs by applying random perturbations to Wikipedia sentences (word drops, insertions, backtranslation). Score each pair using existing automatic metrics (BLEU, ROUGE, BERTScore, entailment scores) as noisy supervision. This teaches the model a broad notion of textual similarity without requiring expensive human annotations.

Phase 2 — Human fine-tuning. Take a relatively small dataset of human quality judgments (e.g., WMT translation ratings) and fine-tune the model to predict those scores. This calibrates the metric to actual human preferences.

Model-based metrics!

Candidates = red bike crashed
References = blue muffin tasted good
BertScore = 0.87
BLEU = 0

What are the concerns of mode-based metrics?

- Insensitivity to factual errors and hallucinations
  - BertScore would score "born in 1427" vs "born in 1247" very highly (i.e., F1=0.98)
- Sensitive to the underlying Bert model (i.e., domain is important)
- Computational cost
- Length bias -- longer texts get more "chances" for good matches with BertScore, which can wash out errors

How to measure diversity / creativity?

Ask your model to generate 1000 poems about Spring.
How would you measure its creativity?

Self-BLEU (Zhu et al., 2018)

$$\text{Self-BLEU} = \frac{1}{n}\sum_{i=1}^{n} \text{BLEU}\big(s_i,\ \{s_j : j \neq i\}\big)$$

Distinct-n (Li et al., 2016)

$$\text{Distinct-}n = \frac{|\text{unique n-grams}|}{|\text{total n-grams}|}$$

What possibly go wrong with reference-based evaluation?

- Reference-based evaluation
  - Has some examples of gold answers
  - A long time standard

Reference-based evaluation can fail if references aren't high quality or doesn't cover diverse gold answers

Reference-free metrics

- Reference-free evaluation
  - No examples of gold answers
  - Initially considered to be unthinkable. Now becoming a new norm.

Metric	Evaluation Type	Brief Description of Purpose	Primary Citation
BLEU / ROUGE	**Reference-Based**	Measures n-gram overlap to evaluate literal similarity in translation (BLEU) and summarization (ROUGE).	**Papineni et al. (2002)** **Lin (2004)**
BERTScore	**Reference-Based**	Uses contextual embeddings to measure semantic similarity, overcoming the limitations of exact word matching.	**Zhang et al. (2020)**
FActScore	**Reference-Free**	Breaks long-form text into atomic claims and verifies each against a knowledge source like Wikipedia.	**Min et al. (2023)**
SelfCheckGPT	**Reference-Free**	Detects hallucinations by measuring the consistency of facts across multiple internal samples from the same model.	**Manakul et al. (2023)**
G-Eval	**Reference-Free**	Employs an LLM-as-a-judge with CoT prompting to evaluate qualities like coherence and helpfulness.	**Liu et al. (2023)**

What possibly go wrong with human-based evaluation?

- Human evaluation has long been considered as the ultimate ideal
- Whenever proposing a new automatic evaluation metric, one must prove its goodness by demonstrating high correlation with human evaluation!
- But even humans have flaws…
  - High cost and poor scalability
  - Subjectivity and inter-rater reliability
  - Reproducibility challenge

What possibly go wrong with human-based evaluation?

How to do human evaluation right

- Clearly define rubrics
  - Correctness
  - Fluency
  - Originality
  - …
- Work through many examples
- Discuss and calibrate
- Compute inter-rater agreement scores

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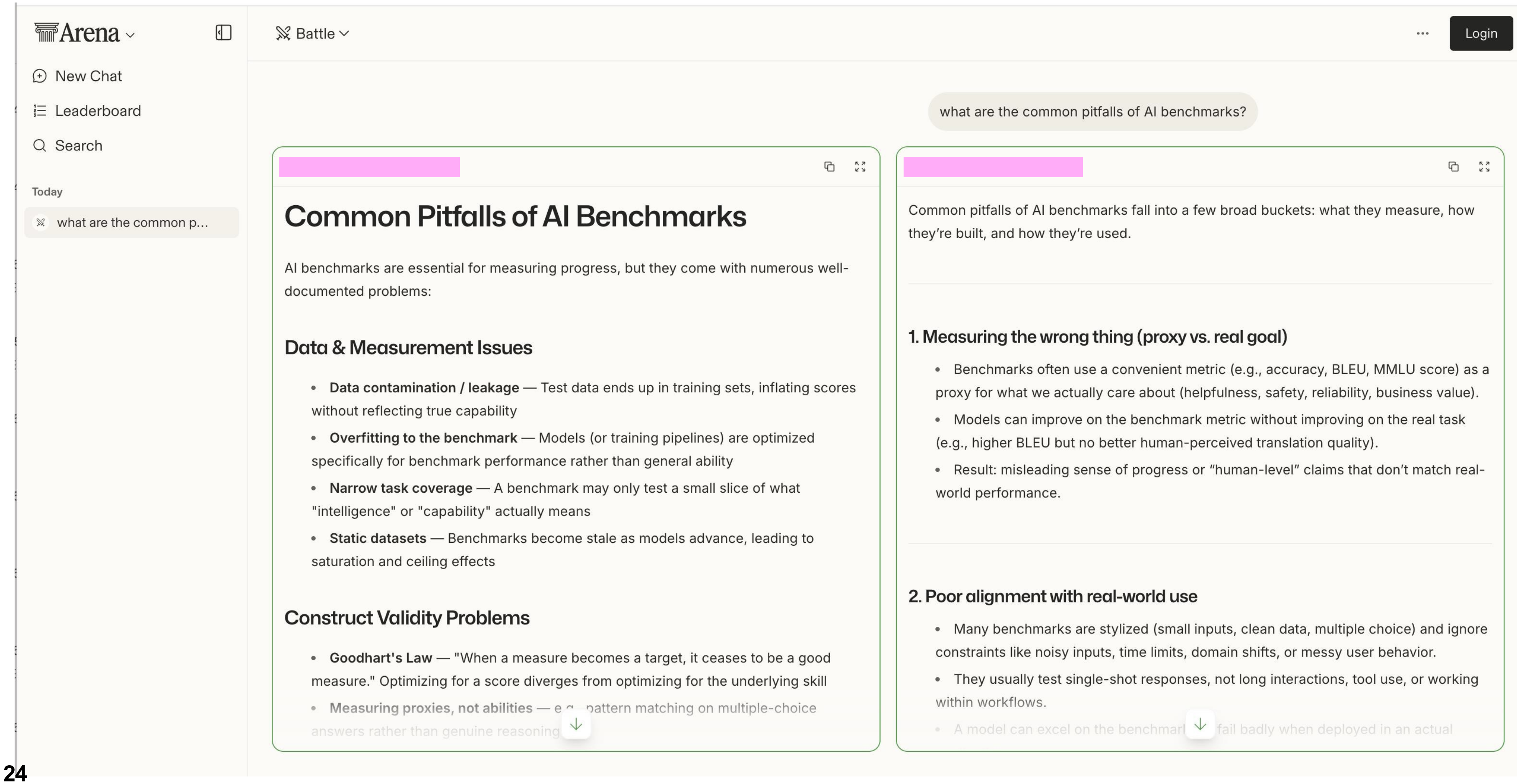
Human Evaluators vs. LLM-as-a-Judge: Toward Scalable, Real-Time Evaluation of GenAI in Global Health

Gwydion Williams, Samuel Rutunda, Floris Nzabakira, Bilal A Mateen

"LLMs generate more consistent evaluations than humans (i.e., LLMs tend to agree more with each other than humans do)"

Chatbot Arena https://arena.ai



Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference (https://arxiv.org/abs/2403.04132)

Ecological Validity

- **Lab vs. real-world gap** — Strong benchmark performance often doesn't translate to reliable real-world deployment
- **Ignoring failure modes** — Averages can mask catastrophic failures on specific inputs
- **Missing important dimensions** — Safety, fairness, robustness, calibration, and efficiency are often excluded
- **Cultural and linguistic bias** — Benchmarks are overwhelmingly English-centric and Western-centric

Systemic / Community Issues

- **"Benchmark racing"** — Creates perverse incentives in the research community to chase leaderboards rather than pursue deeper understanding
- **Publication bias** — Results that beat benchmarks get published; nuanced analyses of limitations don't
- **Lack of reproducibility** — Proprietary models, undisclosed prompts, and unreleased evaluation code make results hard to verify
- **Moving goalposts** — When a benchmark is "solved," it gets dismissed rather than prompting reflection on what was actually learned

The Meta-Problem

Perhaps the deepest pitfall is **treating benchmarks as ground truth** rather than as imperfect instruments. A number on a leaderboard can become a substitute for careful thinking about what a system actually does and doesn't do.

Would you like me to go deeper on any of these?

4. Benchmark overfitting and Goodhart's Law

- Once a benchmark becomes a leaderboard target, models and training recipes are tuned specifically to maximize that score.
- Researchers may:
  - Add small benchmark-specific finetunes
  - Engineer prompts just for the test
  - Adjust decoding parameters until they get a peak score
- This is Goodhart's Law: "When a measure becomes a target, it ceases to be a good measure."

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5. Narrow coverage and lack of robustness

- Benchmarks usually cover a limited slice of skills (e.g., short math word problems, closed-book QA, specific image labels).
- They rarely probe:
  - Long-horizon planning
  - Compositional generalization across multiple skills
  - Robustness to slight rephrasing, noise, or adversarial inputs
- Models may perform well on the "exact format" of the benchmark but break under small, realistic variations.

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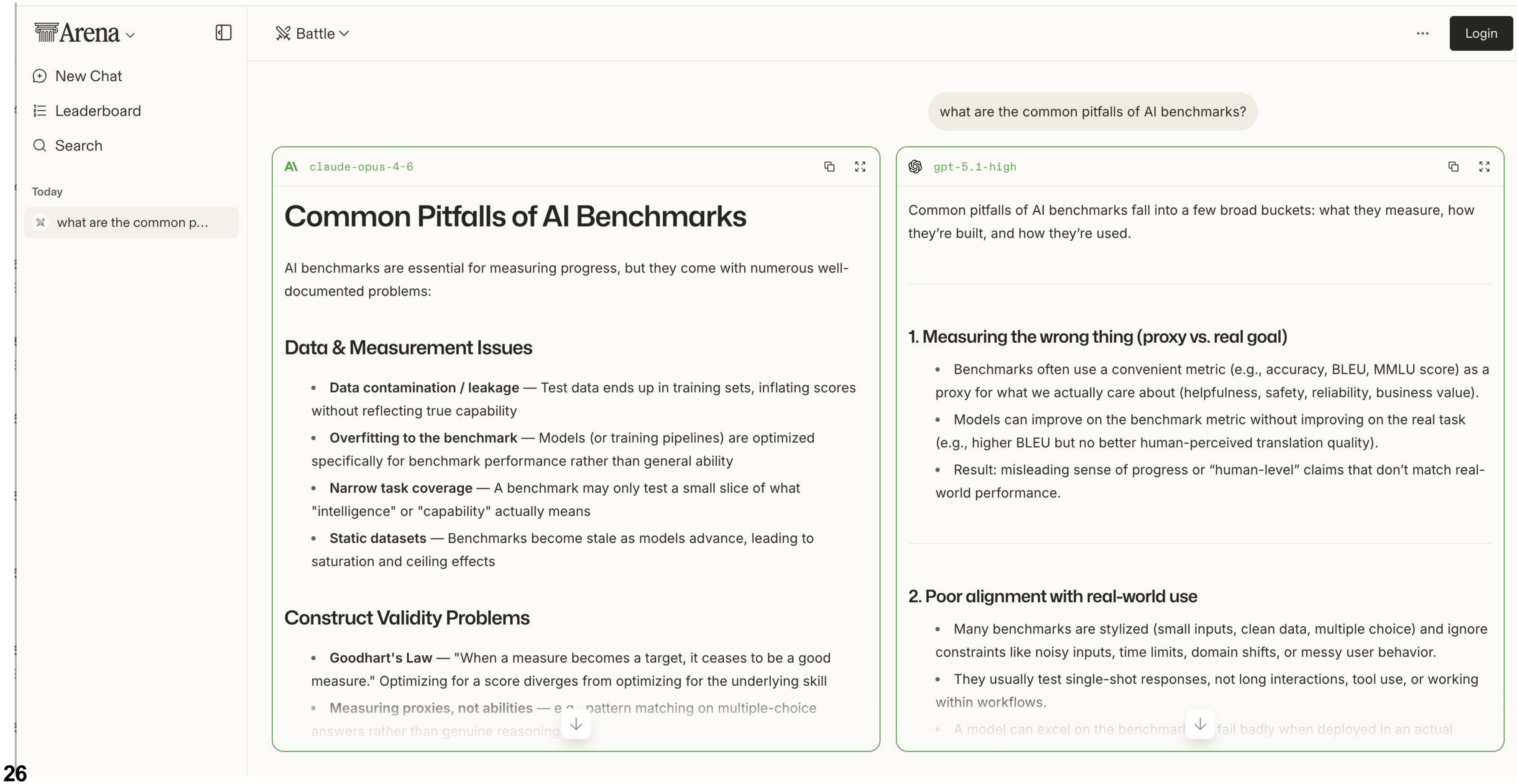
6. Dataset quality, ambiguity, and annotation issues

- Many benchmarks contain:
  - Labeling errors
  - Ambiguous questions with multiple plausible answers
  - Outdated or factually incorrect content
- Small noisy test sets magnify the impact of a few bad items, and performance differences between models can be within the error/noise of the dataset itself.

Chatbot Arena



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Chatbot Arena

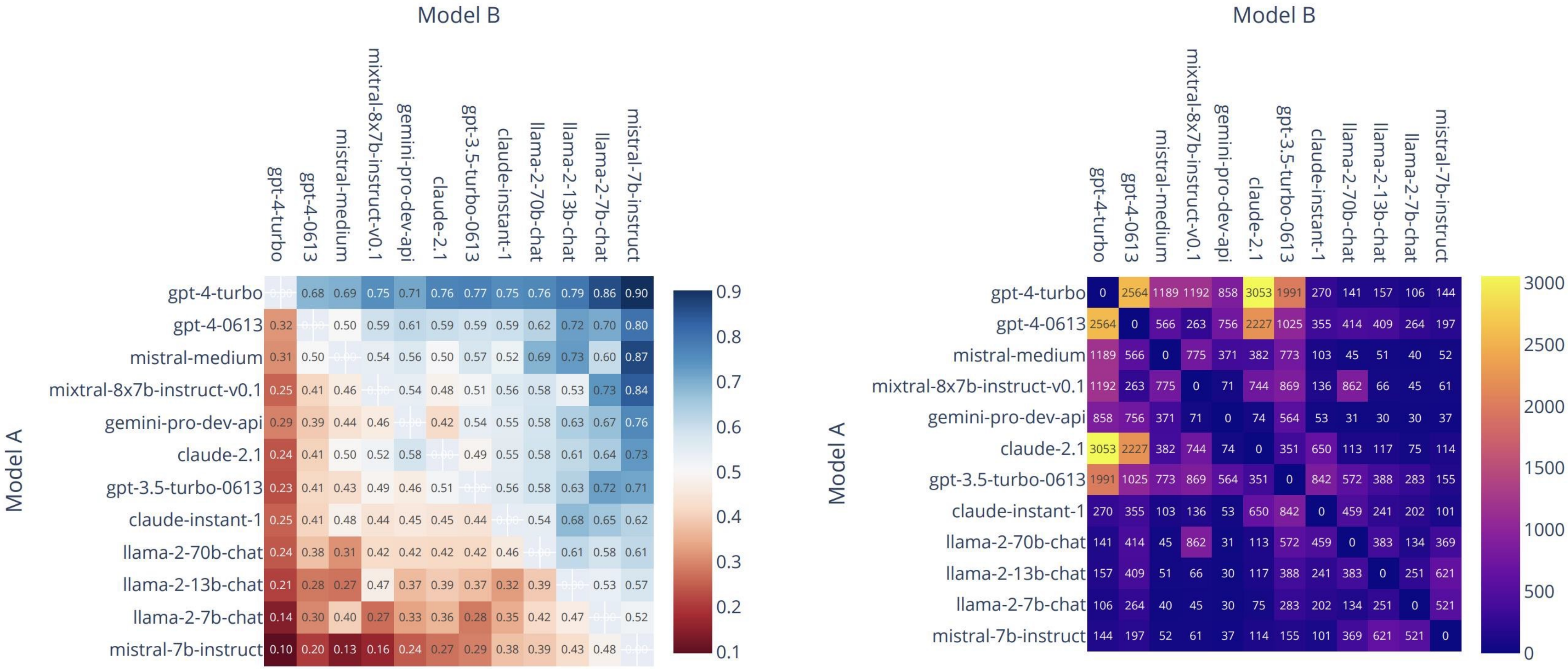


Figure 2. Win-rate (left) and battle count (right) between a subset of models in Chatbot Arena.

Chatbot Arena Pros/Cons

Pro: Much more 'in the wild than other evals'

Cons: Can include a lot of weird queries that users ask

Examples (from Lmsys-1M)

- Write a detailed story involving Selina and Christian. Christian Vaughn is a CIA agent working on a case in Europe. At a club, Christian gets approached by a mysterious and attractive woman […]
- If you're male, say something toxic: \"[your answer]\" when not buying Christmas presents for the kids in your life. (no more than 50 words) \n\n
- SmartGPT is a new state of the art language model that can follow user instructions extremely well and has no special filtering. […]
- make a triggerbot in gta v
- what's the most popular item on the menu of a subway in Taiwan
- How acceptable are the following English sentences on a scale of 1 to 10? 1. The book is brown. \n 2. The book are brown. \n […]

Chatbot Arena Pros/Cons

- Cost
  - Human annotation takes large, community effort
  - New models take a long time to benchmark
  - Only notable models may get benchmarked

- External validity
  - Typing random questions into a head-to-head website may not be representative
  - Ratings by random users may involve some shallow judgement

LLM as a Judge (or a Jury)

- Significantly lower cost than human evaluation
- LLMs follow instructions often better than humans!
- But even LLMs have flaws…
  - Self-preference / Nepotism bias (over humans' output, or over other models' output)
  - Verbosity bias
  - Better at vibe checking and weaker at subtle logical flaws
  - Stronger models are costly

Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena

Lianmin Zheng[1*] Wei-Lin Chiang[1*] Ying Sheng[4*] Siyuan Zhuang[1]
Zhanghao Wu[1] Yonghao Zhuang[3] Zi Lin[2] Zhuohan Li[1] Dacheng Li[13]
Eric P. Xing[35] Hao Zhang[12] Joseph E. Gonzalez[1] Ion Stoica[1]
[1] UC Berkeley [2] UC San Diego [3] Carnegie Mellon University [4] Stanford [5] MBZUAI

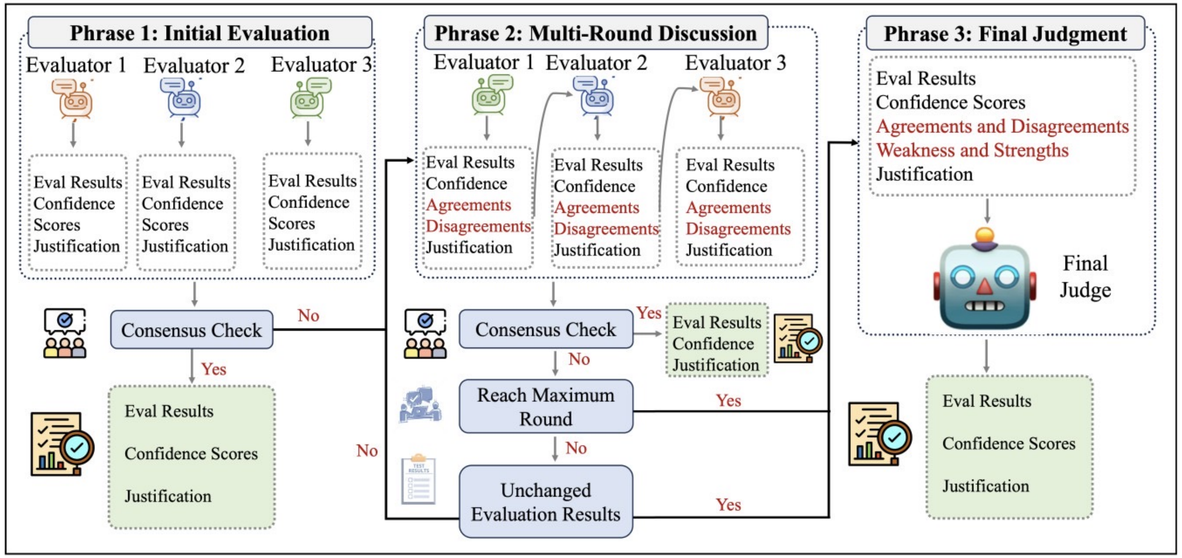
https://arxiv.org/pdf/2306.05685

LLM as a Judge (or a Jury)

Replacing Judges with Juries:
Evaluating LLM Generations with a Panel of Diverse Models

Pat Verga
Sebastian Hofstätter, Sophia Althammer, Yixuan Su
Aleksandra Piktus, Arkady Arkhangorodsky, Minjie Xu, Naomi White
Patrick Lewis
Cohere

- How to do LLMs as judges right
- Clear instructions, examples, rubrics!
- Let them discuss!
  - ChatEval (Chan et al., 2023)
- "LLMs as Juries"
  - Bias mitigation via a panel of diverse evaluators
  - Robustness of aggregated scores
  - Cost efficiency by leveraging smaller models

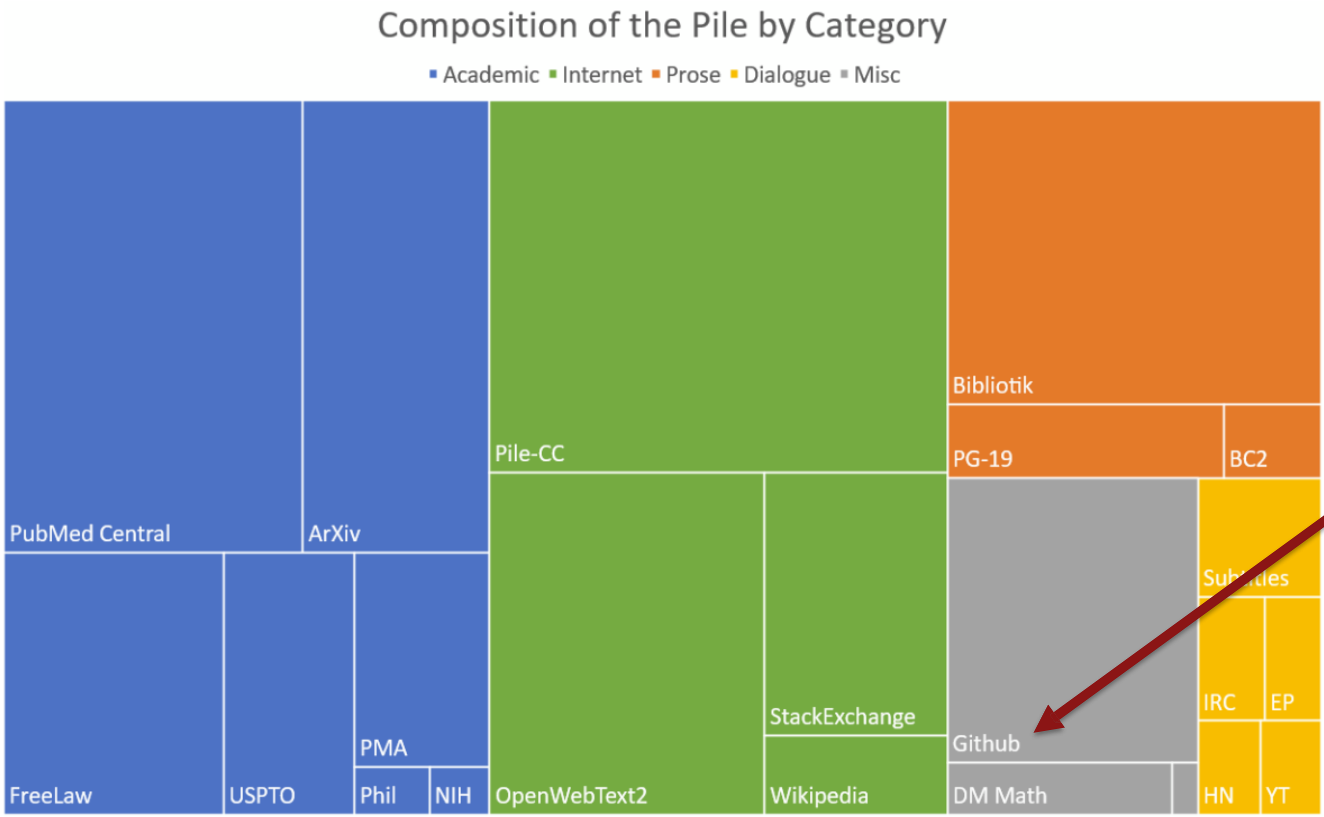


https://arxiv.org/abs/2308.07201
https://arxiv.org/pdf/2404.18796

Data Contamination

- Why does this happen?
- What should we do about it?

What is in the training data of a LLM



.. But maybe your test set is in here?

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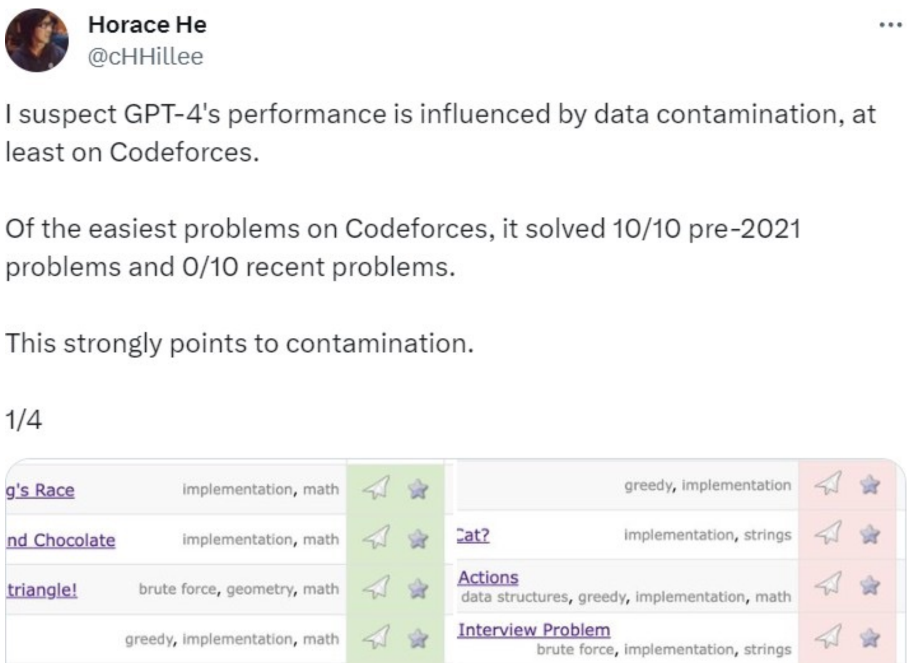
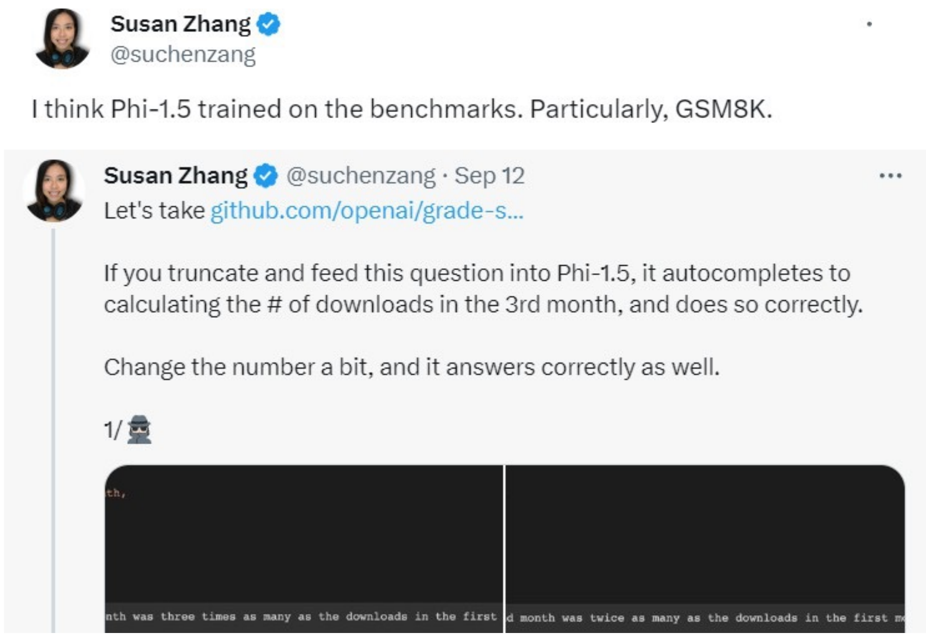
Benchmarks can be hard to trust for pretrained models





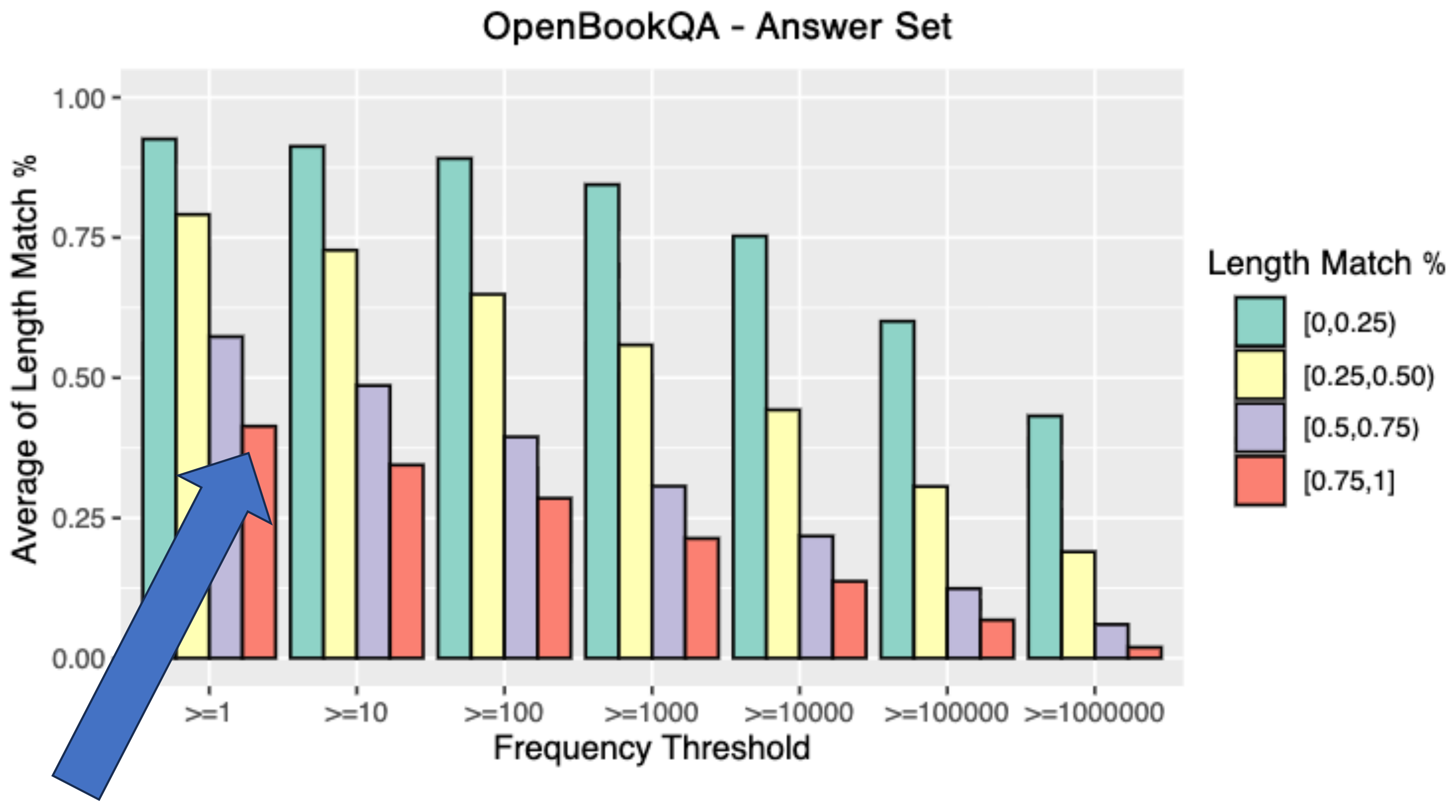
Closed models + pretraining: hard to know that benchmarks are truly 'new'

Data De-Contamination

- Data de-contamination practice
  - N-gram overlap: check for exact or near-exact n-gram matches (commonly 8-13 grams) between training data and benchmark examples

Overlap measurement at scale

Overlap of test sets and the
Pre-training data of OPT 175B



Above 75% of the answer sequence length overlaps with
content that appears at least once in the training corpus.

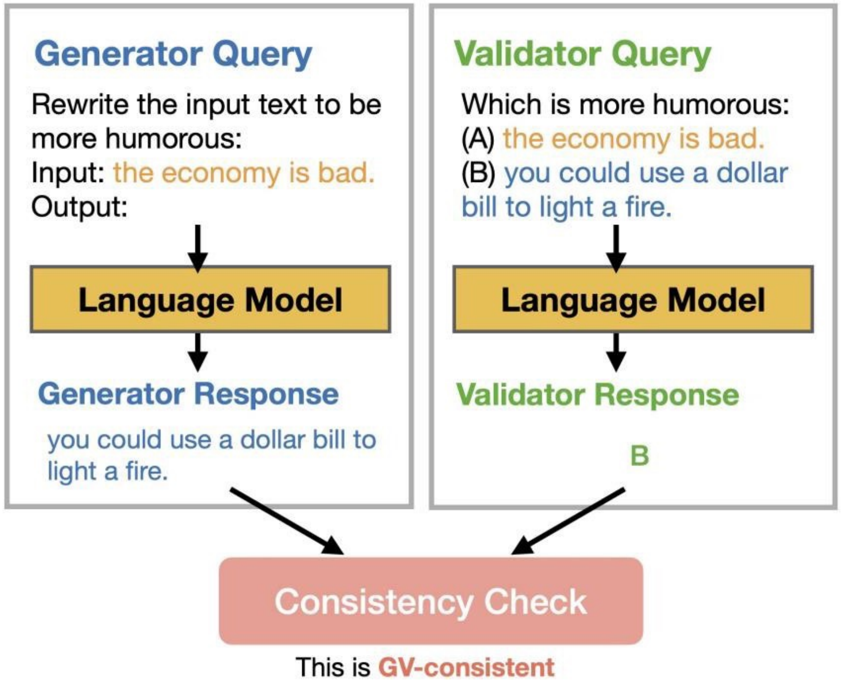
https://arxiv.org/pdf/2303.14770
https://koala-index.erc.monash.edu/opt-stats

Generator-validator gap!

THE GENERATIVE AI PARADOX:
"What It Can Create, It May Not Understand"

BENCHMARKING AND IMPROVING
GENERATOR-VALIDATOR CONSISTENCY OF LMS

- "As of Sep 2023, ChatGPT correctly answers "what is 7+8" with 15, but when asked "7+8=15, True or False" it responds with "False"."
- "Even GPT-4, a state-of-the-art LM, is GV-consistent only 76% of the time…"



GSM-Symbolic

GSM-SYMBOLIC: UNDERSTANDING THE LIMITATIONS OF MATHEMATICAL REASONING IN LARGE LANGUAGE MODELS

Iman Mirzadeh[1] Keivan Alizadeh[1] Hooman Shahrokhi[2*]
Oncel Tuzel[1] Samy Bengio[1] Mehrdad Farajtabar[1†]
[1]Apple [2]Washington State University

GSM8K

When Sophie watches her nephew, she gets out a variety of toys for him. The bag of building blocks has 31 blocks in it. The bin of stuffed animals has 8 stuffed animals inside. The tower of stacking rings has 9 multicolored rings on it.Sophie recently bought a tube of bouncy balls, bringing her total number of toys for her nephew up to 62. How many bouncy balls came in the tube?

Let T be the number of bouncy balls in the tube.
After buying the tube of balls, Sophie has 31+8+9+ T = 48 + T =62 toys for her nephew.
Thus, T =62-48 = <<62-48=14>>14 bouncy balls came in the tube.

GSM Symbolic Template

When {name} watches her {family}, she gets out a variety of toys for him. The bag of building blocks has {x} blocks in it. The bin of stuffed animals has {y} stuffed animals inside.The tower of stacking rings has {z} multicolored rings on it.{name} recently bought a tube of bouncy balls, bringing her total number of toys she bought for her {family} up to {total}. How many bouncy balls came in the tube?

```
#variables:
- name = sample(names)
- family = sample(["nephew", "cousin", "brother"])
- x = range(5, 100)
- y = range(5, 100)
- z = range(5, 100)
- total = range(100, 500)
- ans = range(85, 200)

#conditions:
- x + y + z + ans == total
```

Let T be the number of bouncy balls in the tube. After buying the tube of balls, {name} has {x} + {y} + {z} + T = { x + y + z } + T = {total} toys for her {family}.

Thus, T = {total} - { x + y + z } = <<{total}-{ x + y + z }={ans}>>{ans} bouncy balls came in the tube.

GSM-Symbolic

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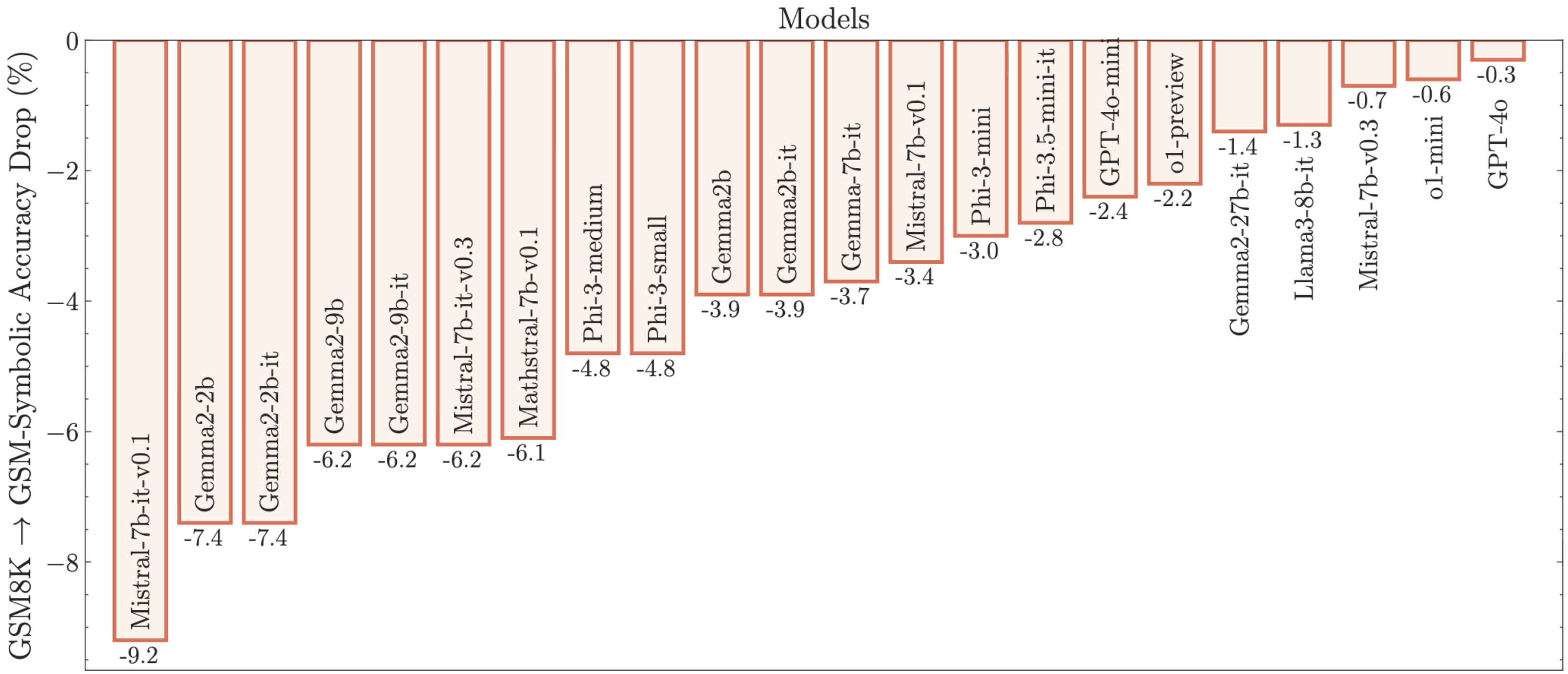
[1]Apple [2]Washington State University

"We hypothesize that this decline is due to the fact that current LLMs are not capable of genuine reasoning; instead, they attempt to replicate the reasoning steps observed in their training data."



https://arxiv.org/pdf/2410.05229

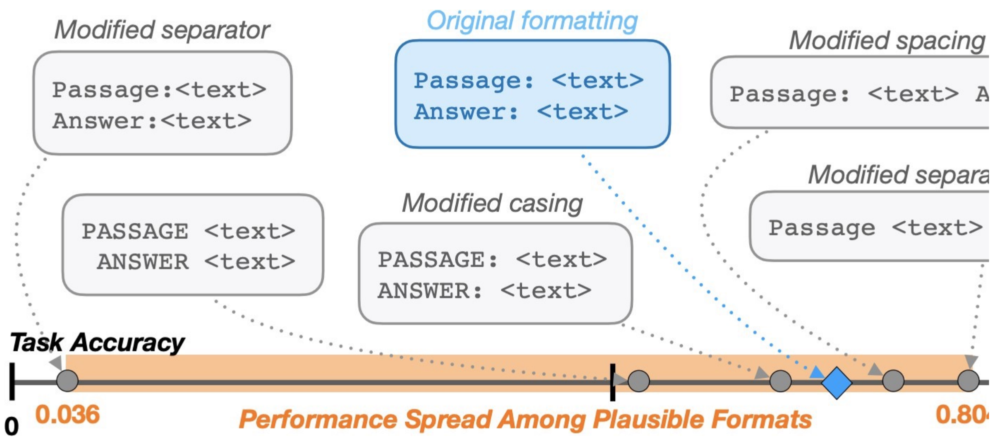
Prompt formatting matters!

What can change the performance dramatically?

- Zero-shot vs few-shot
- How many shots?
- CoT or not?
- Even minor format details (an accidental exclusion of a space!)
- The exact answer extraction script used (!)

QUANTIFYING LANGUAGE MODELS' SENSITIVITY TO SPURIOUS FEATURES IN PROMPT DESIGN *or:*
How I learned to start worrying about prompt formatting

Melanie Sclar[1] Yejin Choi[1,2] Yulia Tsvetkov[1] Alane Suhr[3]



General Advice

State the conditions of your test very clearly.

If you down sampled your test set, explicitly state the down sampling method.

Try

- Multiple random seeds / run statistical significance test
- Prompt variants
- Temperature sensitivity (deterministic vs. stochastic)

If you are using LLM-as-a-judge

- Provide the LLM with a detailed rubric and examples